CS 4210 – Assignment #4

Maximum Points: 100 pts.

Bronco ID: |__|__|__|__|__|__|__|__|__|

Last Name: _____

First Name: _____

Note 1: Your submission header must have the format as shown in the above-enclosed rounded rectangle.
Note 2: Homework is to be **done individually**. You may discuss the homework problems with your fellow students, but you are NOT allowed to copy – either in part or in whole – anyone else's answers.
Note 3: Your deliverable should be a **.pdf file** submitted through **Gradescope** by the deadline. Do not forget to **assign a page to each of your answers** when making a submission. In addition, source code (.py files) should be added to an **online repository (e.g., GitHub)** to be downloaded and executed later.
Note 4: All submitted materials must be legible. Figures/diagrams must be of good quality.
Note 5: Please use and check the Canvas discussion for further instructions, questions, answers, and hints. The bold words/sentences provide information for a complete or accurate answer.

1. [30 points]. Given an input image with 3 color channels (Red, Green, and Blue), each represented as a 4×4 matrix, answer the following questions.

   a) [25 points]. Perform a 3D convolution using **no padding**, a **stride of 1**, and apply the **ReLU activation** function to the resulting values to generate an activation map. Hint: perform an element-wise multiplication between each 2×2 kernel slice and its corresponding 2×2 region in the input channels, sum over all channels, and produce a single scalar output before applying ReLU - Activation(i,j) = ReLU(RedSum + GreenSum + BlueSum).

   - Red Channel

$$C = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 3 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 \\ 2 & 1 & 0 & 0 \end{bmatrix}$$

   - Green Channel

$$C = \begin{bmatrix} 0 & 1 & 2 & 1 \\ 2 & 2 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 2 & 1 & 1 \end{bmatrix}$$

   - Blue channel

$$C = \begin{bmatrix} 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 \\ 0 & 2 & 1 & 1 \\ 2 & 0 & 0 & 1 \end{bmatrix}$$

   - Red Kernel

$$K = \begin{bmatrix} 0 & 0 \\ -1 & 2 \end{bmatrix}$$

   - Green Kernel

$$K = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

   - Blue kernel

$$K = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

   b. [5 points] Perform an average pooling operation with a **2×2 filter** and a **stride of 1** on the activation map obtained in **part (a)**.

2. [25 points]. You are given a small dataset consisting of labeled and unlabeled examples. Each instance has two features: Color and Shape. Your task is to train a classifier to predict the test samples iteratively using Self-Training. Requirement: Use **Naïve-Bayes** with **m-estimate** (m=1, p=0.5, p(A)=0.3, and p(B)=0.7) as your classifier and expand your training set by pseudo-labeling the unlabeled data predicted with **confidence ≥ 90%**. Calculate the performance of your **baseline** and **self-trained** classifier for comparison purposes, showing the individual predictions used to compute each accuracy. **Only one iteration** is needed for Self-Training. Show your math for full marks.

Original: $P(A_i|C) = \frac{N_{ic}}{N_c}$ m-estimate: $P(A_i|C) = \frac{N_{ic}+mp}{N_c+m}$

Initial Labeled Data (Training Set)

ID	Color	Shape	Label
1	Red	Circle	A
2	Red	Square	A
3	Blue	Circle	B

Unlabeled Data

ID	Color	Shape
4	Blue	Square
5	Blue	Circle
6	Red	Circle
7	Blue	Square

Labeled Data (Test Set)

ID	Color	Shape	Label
8	Blue	Square	B
9	Red	Circle	A
10	Red	Square	A
11	Blue	Circle	B

3. [25 points]. Given the small dataset relating temperature (°C) to ice cream sales (units sold) provided, answer the questions below.

Temperature	Sales
10	5
12	6
15	20
18	22
25	24
30	10

   a) [20 points]. Identify the **threshold γ** that should be used as the root (first split) of the regression tree. Show all your **calculations for SSR**.
   b) [5 points]. Plot the **depth-1 regression tree** with its corresponding predictions.

4. [20 points]. Complete the Python program (cnn.py) that will train a Convolutional Neural Network (CNN) using TensorFlow and Keras to classify 32x32 images of handwritten digits (0 through 9). Your task is to 1) load the dataset using the provided helper function, 2) normalize and reshape the input data for the CNN, 3) construct a CNN architecture following the guidelines in the code, 4) train the model on the training set and validate on the test set, 5) evaluate the final model and 6) print the test accuracy.

Important Note: Answers to all questions should be written clearly, concisely, and unmistakably delineated. You may resubmit multiple times until the deadline (the last submission will be considered).

NO LATE ASSIGNMENTS WILL BE ACCEPTED. ALWAYS SUBMIT WHATEVER YOU HAVE COMPLETED FOR PARTIAL CREDIT BEFORE THE DEADLINE!